

Dokumentasi Analisis DEG



**TUGAS MINGGU
KE- 2
PROJECT BRSP**

Presented by

**CANTIKA CHINTYA
LUTHFIANNISSA, S. Biotek**



NCBI > GEO > **Accession Display** [?](#) Not logged in | [Login](#) [?](#)

Scope: Self Format: HTML Amount: Quick GEO accession: GSE52889 GO

Series GSE52889

[Query DataSets for GSE52889](#)

Status	Public on Sep 05, 2014
Title	Effect of methylphenidate on gene expression profiles of Attention Deficit Hyperactivity Disorder patients and unaffected controls.
Organism	Homo sapiens
Experiment type	Expression profiling by array
Summary	Attention deficit hyperactivity disorder (ADHD) is a common psychiatric condition of children with a prevalence of 5-10% worldwide. Up to 30% of adults with a history of childhood ADHD maintain symptoms in later life; these adult ADHD patients are severely impaired in social and professional life due to persistence of ADHD core symptoms like impulsivity, attention deficit and hyperactivity as well as frequently observed co-morbidities like alcohol and drug abuse, major depression, bipolar and personality disorders. Pharmaceutical treatment options include methylphenidate (MPH), which is amongst others an inhibitor of the dopamine transporter and therefore increases dopamine levels in the brain. However, not all ADHD patients are MPH responders with clinical features to distinguish responders and non-responders being not at hand so far. Likewise, neurobiological reasons for drug response are still elusive.
Overall design	Here, we examined the global transcriptional response of MPH on lymphoblastoid cell lines (LCLs) derived from ADHD patients and unaffected controls.
Contributor(s)	Schwarz R , Scholz CJ , Weber H , Reif A , Kittel-Schneider S
Citation(s)	Schwarz R, Reif A, Scholz CJ, Weissflog L et al. A preliminary study on methylphenidate-regulated gene expression in lymphoblastoid cells of ADHD patients. <i>World J Biol Psychiatry</i> 2015 Apr;16(3):180-9. PMID: 25162476

Analyze with GEO2R

Submission date: Dec 02, 2013

Use GEO2R to compare two or more groups of Samples in order to identify genes that are differentially expressed across experimental conditions. Results are presented as a table of genes ordered by significance. [Full instructions](#) [YouTube](#)

GEO accession

GSE52889

Set

Effect of methylphenidate on gene expression profiles of Attention Deficit Hyperactivity Disorder patients and unaffected controls.

▼ Samples

▸ Define groups

Selected 4 out of 4 samples

Group	Accession	Title	Source name	Affection	Treatment	Tissue	Pool size
untreated	GSM1277538	ctrl_untreat	RNA pool from untreated LCLs generated from unaffected controls	unaffected	untreated	EBV-transformed lymphoblastoid cell lines	LCLs from 6 unaffected controls
untreated	GSM1277537	ADHD_untreat	RNA pool from untreated LCLs generated from ADHD patients	ADHD	untreated	EBV-transformed lymphoblastoid cell lines	LCLs from 7 ADHD patients
treated	GSM1277540	ctrl_MPH	RNA pool from MPH-treated LCLs generated from unaffected controls	unaffected	MPH	EBV-transformed lymphoblastoid cell lines	LCLs from 6 unaffected controls
treated	GSM1277539	ADHD_MPH	RNA pool from MPH-treated LCLs generated from ADHD patients	ADHD	MPH	EBV-transformed lymphoblastoid cell lines	LCLs from 7 ADHD patients

GEO2R

Options

Profile graph

R script

▼ Quick start

- Specify a GEO Series accession and a Platform if prompted.
- Click 'Define groups' and enter names for the groups of Samples you plan to compare, e.g., test and control.
- Assign Samples to each group. Highlight Sample rows then click the group name to assign those Samples to the group. Use the Sample metadata (title, source and characteristics) columns to help determine which Samples belong to which group.
- Click 'Analyze' to perform the calculation with default settings.
- You may change settings in the Options tab.

How to use

Analyze

[Full instructions](#) [YouTube](#)

GEO accession	GSE52889	Set
----------------------	----------	-----

Selected 4 out of 4 samples

GEO2R	Options	Profile graph	R script
-------	---------	---------------	----------

☒ Benjamini & Hochberg (False discovery rate)

☐ Benjamini & Yekutieli

☐ Bonferroni

☐ Holm

☒ Auto-detect

☐ Yes

☐ No

☐ Yes

☒ No

☐ Yes

☒ No

☐ Submitter supplied

☒ NCBI generated

Significance level cut-off
(enter number between 0 and 1)

Log 2 fold change threshold	0.5
-----------------------------	-----

1 selected (clear)

- ✔ untreated vs treated

Use GEO2R to compare two or more groups of Samples in order to identify genes that are differentially expressed across experimental conditions. Results are presented as a table of genes ordered by significance. [Full instructions](#) [YouTube](#)

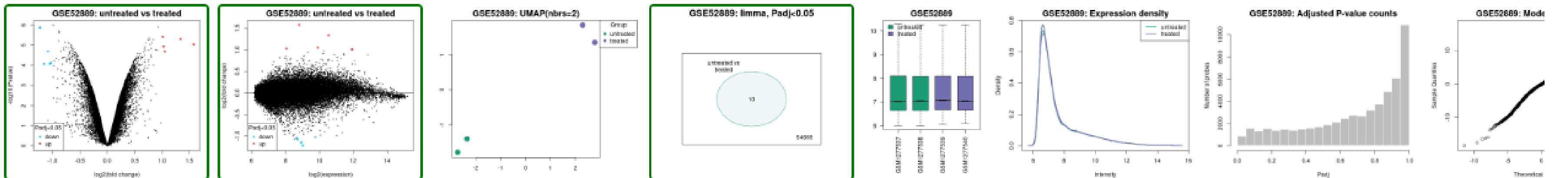
GEO accession **Set** [Effect of methylphenidate on gene expression profiles of Attention Deficit Hyperactivity Disorder patients and unaffected controls.](#)

[▶ Samples](#) [▶ Define groups](#) Selected 4 out of 4 samples

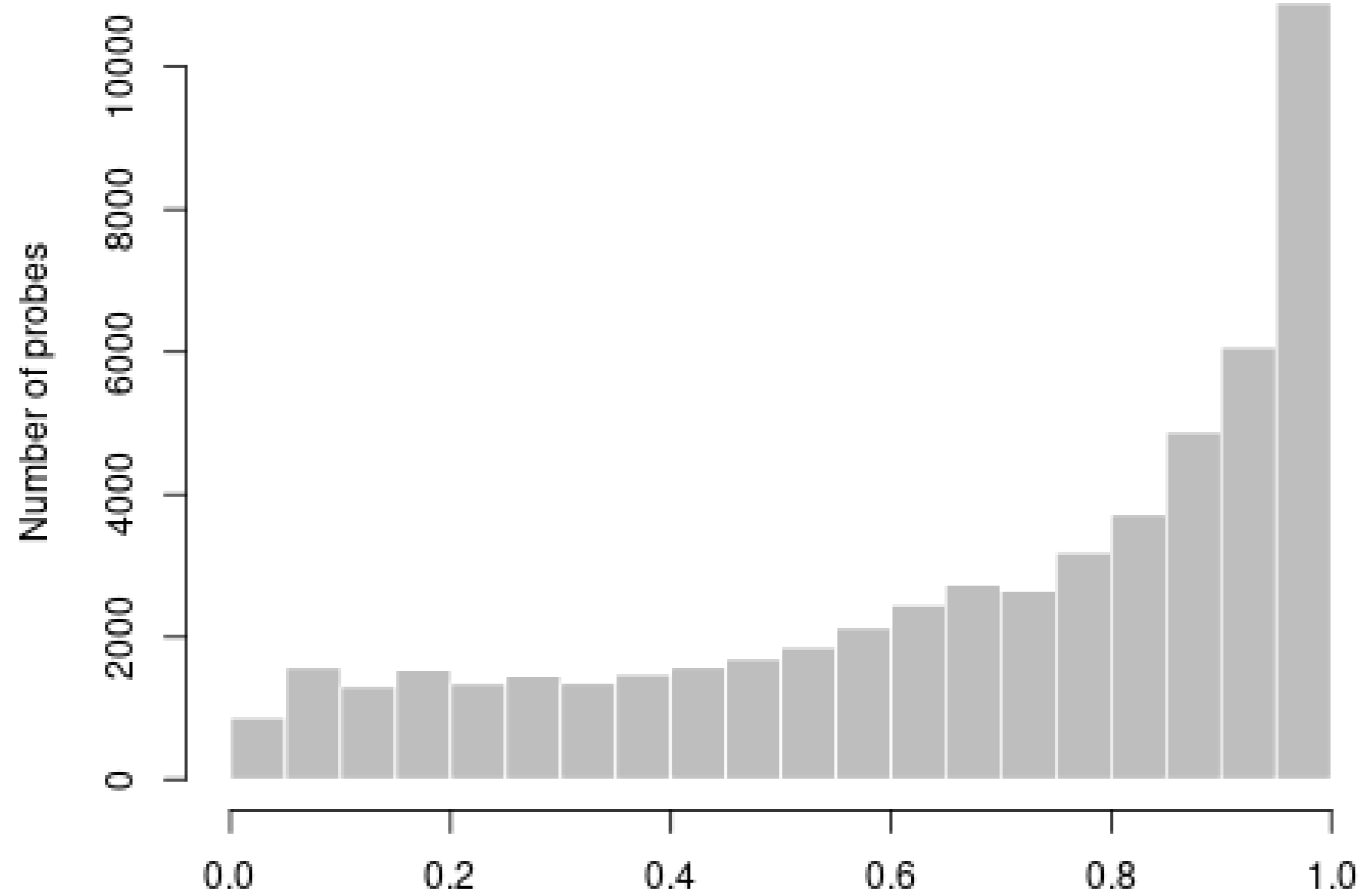
[GEO2R](#) [Options](#) [Profile graph](#) [R script](#)

[Reanalyze](#) if you changed any options.

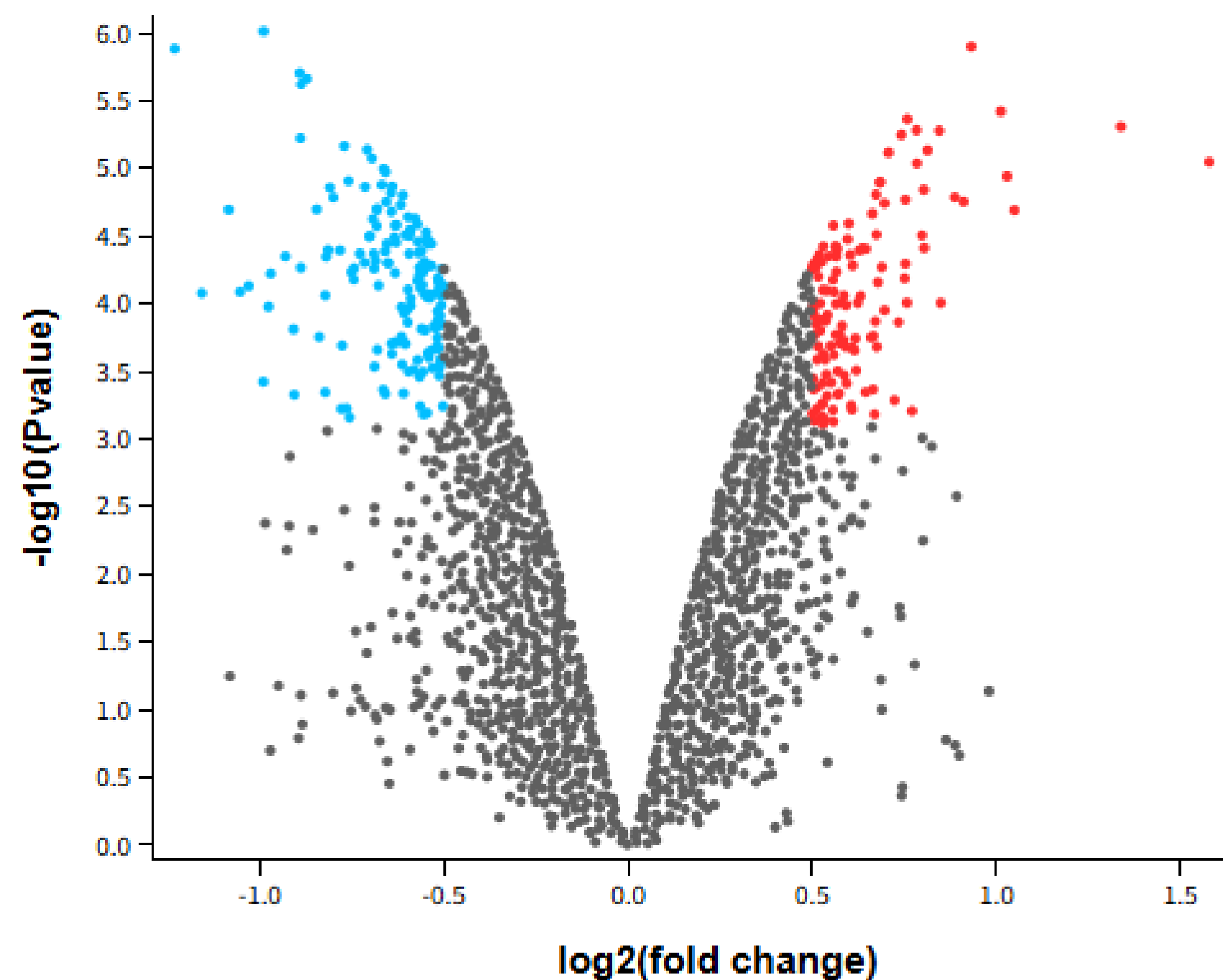
Visualization [?](#)



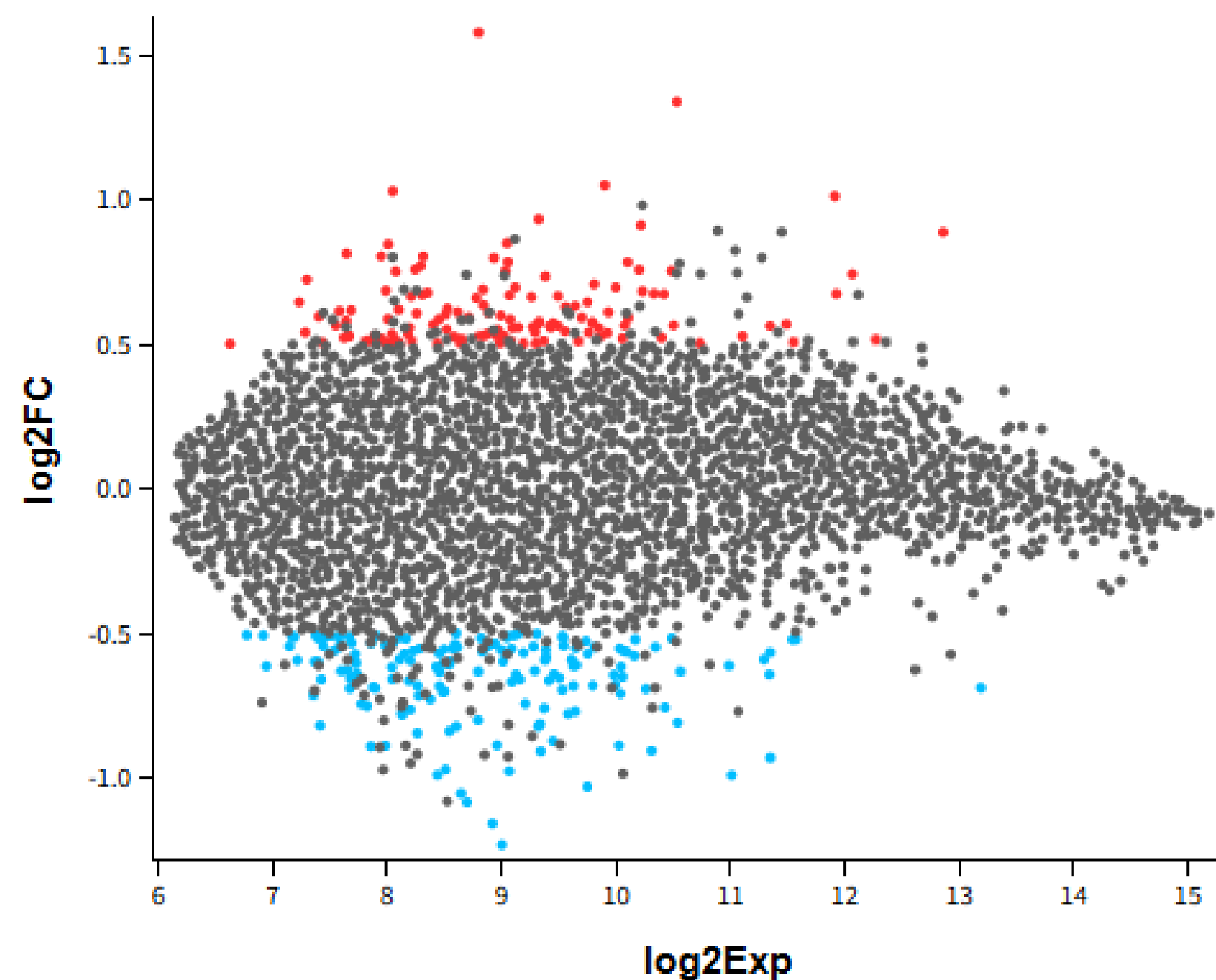
GSE52889: Adjusted P-value counts



Volcano plot
GSE52889: Effect of methylphenidate on gene expression profiles of...
untreated vs treated, $P_{adj} < 0.05$



Meandiff plot
GSE52889: Effect of methylphenidate on gene expression profiles of...
untreated vs treated, $P_{adj} < 0.05$



Top differentially expressed genes

[Download full table](#) [Select columns](#)

ID	adj.P.Val	P.Value	t	B	logFC	Gene.symbol	Gene.title
▶ 213359_at	0.0217	9.65e-07	-18.44	5.6	-0.989	HNRNPD	heterogeneous nuclear rib...
▶ 227936_at	0.0217	1.25e-06	17.69	5.45	0.932	TMEM68	transmembrane protein 68
▶ 209007_s_at	0.0217	1.30e-06	-17.58	5.42	-1.231	RSRP1	arginine and serine rich pr...
▶ 204621_s_at	0.0217	1.97e-06	-16.44	5.16	-0.891	NR4A2	nuclear receptor subfamily...
▶ 239413_at	0.0217	2.16e-06	-16.2	5.1	-0.872	CEP152	centrosomal protein 152
▶ 229434_at	0.0217	2.38e-06	-15.95	5.04	-0.887		
▶ 204103_at	0.0241	3.78e-06	14.8	4.73	1.012	CCL4	C-C motif chemokine ligand...
▶ 231863_at	0.0241	4.31e-06	14.49	4.63	0.758	ING3	inhibitor of growth family ...
▶ 203665_at	0.0241	4.89e-06	14.19	4.54	1.338	HMOX1	heme oxygenase 1
▶ 228617_at	0.0241	5.19e-06	14.06	4.5	0.783	XAF1	XIAP associated factor 1
▶ 1554878_a_at	0.0241	5.23e-06	14.04	4.49	0.845	ABCD3	ATP binding cassette subf...
▶ 203380_x_at	0.0241	5.63e-06	13.88	4.44	0.742	SRSF5	serine and arginine rich sp...
▶ 235716_at	0.0241	5.94e-06	-13.75	4.4	-0.889		
▶ 202220_at	0.0241	6.80e-06	-13.45	4.3	-0.77	KIAA0907	KIAA0907
▶ 211996_s_at	0.0241	7.23e-06	-13.32	4.25	-0.708	LOC101060275//NPIPA5/...	putative NPIP-like protein ...
▶ 1559249_at	0.0241	7.32e-06	13.3	4.24	0.813	ATXN1	ataxin 1
▶ 238935_at	0.0241	7.60e-06	13.21	4.21	0.707	RPS27L	ribosomal protein S27 like
▶ 225318_at	0.0241	8.38e-06	-13	4.14	-0.695	WHSC1L1	Wolf-Hirschhorn syndrome...
▶ 202859_x_at	0.0241	8.91e-06	12.88	4.09	1.578	CXCL8	C-X-C motif chemokine lig...
▶ 227650_at	0.0241	9.15e-06	12.82	4.07	0.784	HSPA14	heat shock protein family ...
▶ 225957_at	0.0241	1.00e-05	-12.63	4	-0.663	CREBRF	CREB3 regulatory factor
▶ 230256_at	0.0241	1.06e-05	-12.52	3.96	-0.658	RUSC1-AS1	RUSC1 antisense RNA 1
▶ 214963_at	0.0241	1.14e-05	12.36	3.9	1.029	NUP160	nucleoporin 160
▶ 228098_s_at	0.0241	1.23e-05	-12.21	3.84	-0.759	MYLIP	myosin regulatory light ch...
▶ 228488_at	0.0241	1.26e-05	-12.17	3.83	-0.685	ZC3AN2	zinc finger and domain...

The background features a light blue globe with a grid pattern, centered behind the text. Surrounding the globe are several large, solid blue circles of varying sizes. Some of these circles have a halftone dot pattern on their edges, creating a layered, modern aesthetic.

THANK YOU